

PAPER_894: SCm E(t) Lagrangian Variation

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Abstract

SCm-specific E(t) Lagrangian variation in the superconductive vacuum. $L_{SCm} = \rho_{SCm}(t) \cdot V \cdot c^2 \cdot (2R-1) \cdot V_{fil} \cdot S_{26}$ with Euler-Lagrange variation $\delta S / \delta \varphi_{SCm} = 0$ providing the SCm closure equation.

1. Core Equations

$$L_{SCm} = E_{net,SCm} \cdot V \cdot S_{26}$$
$$\delta S / \delta \varphi_{SCm} = 0$$

2. Parameters

Parameter	Default	Description
t	0.0 s	Time parameter
V_filament	1e48 m ³	Filament volume
F_UBi_over_FU	0.8	Buoyancy-to-field ratio

3. UQFF Integration

This calculator integrates et_scm_vacuum.py from Session 207 into the CP4 pipeline as class #478.

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